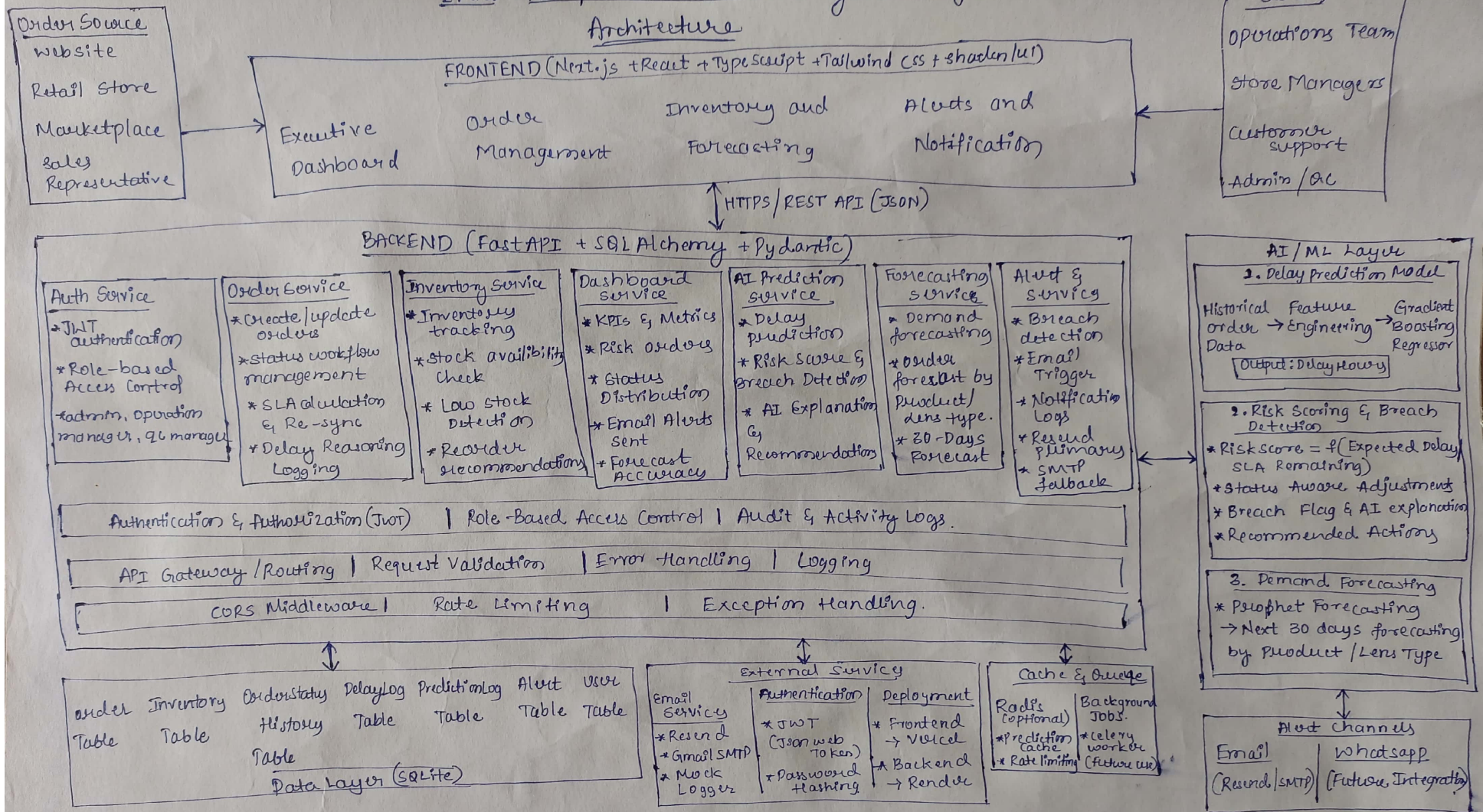
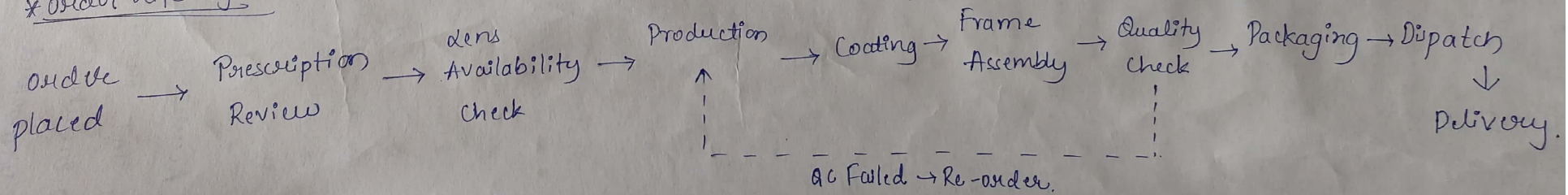


Fluro - AI powered Order Management System

Architecture



* Order life Cycle :-



Algorithms Used

Machine Learning Algorithm:

a) Gradient Boosting Regressor (skit-learn)

- * Model $\rightarrow$ sklearn.ensemble.GradientBoostingRegressor
- * Def $\rightarrow$ Ensemble of decision trees that predicts continuous numeric values.
- * Purpose $\rightarrow$ It is used to predict turnaround time (TAT) in hours.

b) Gradient Boosting Classifier (skit-learn)

- * Model $\rightarrow$ sklearn.ensemble.GradientBoostingClassifier
- * Def $\rightarrow$ Ensemble of decision tree that predicts binary class outcomes.
- * Purpose $\rightarrow$ Predicts SLA breach probability and compute risk score.

c) Facebook Prophet

- * Model $\rightarrow$ prophet.Prophet
- * Def $\rightarrow$ Time series forecasting library for trend and seasonality.
- * Purpose $\rightarrow$ Forecast inventory demand for 7 to 30 days.

Fallback/Heuristic Algorithms:

- a) Rule based prediction $\rightarrow$ Deterministic scoring using business logic factors.
- * used for forecasting predictor when machine learning models are unavailable.

Data Preprocessing and validation:

a) One-hot encoding $\rightarrow$ Converts categories into binary numeric columns.

- * used for categorical features: store_location, lens_type, coating - etc.

b) Standard scaling $\rightarrow$ Normalizes numerical features to zero mean & unit variance.

- * used for numerical features: power, lens_index, sla_hours - etc

c) Column Transformer Pipeline $\rightarrow$ used for combining categorical & numerical preprocessing.

d) Train-Test Split $\rightarrow$ Divides data into training and testing sets.

- * used for validating model performance on unseen data.

e) Gridsearch cv $\rightarrow$ Searches all combination of hyperparameters to find the best model.

- * used to optimize model accuracy & prevent overfitting.

Security & Other Algorithms:

a) JINT Authentication with HMAC-SHA256 $\rightarrow$ Signed token-based authentication.

- * used for secure login & protect API routes.

b) SLA Risk Scoring Heuristic $\rightarrow$ Converts predicted delay into a 0-100 %.

- * used to visualize order risk with color-coded badges.

Accuracy of the Services:-

$$\text{① TAT prediction } R^2 \text{ score} = 0.9400 \text{ (if 1, perfect model)} \rightarrow \left[R^2 = 1 - \frac{SS_{\text{result}}}{SS_{\text{total}}} \right]$$

$$\text{② Breach classifier Accuracy} = 1.0000$$

$$\text{③ Dashboard forecast accuracy} = 92\%$$